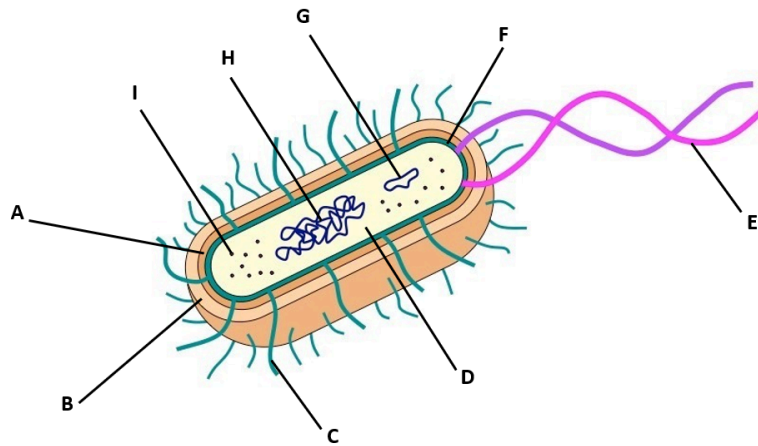


## Cell Study Guide for Honors Biology

Use this study guide to help you prepare for your test. You may write your answers on a separate sheet of paper. It is NOT for a grade.

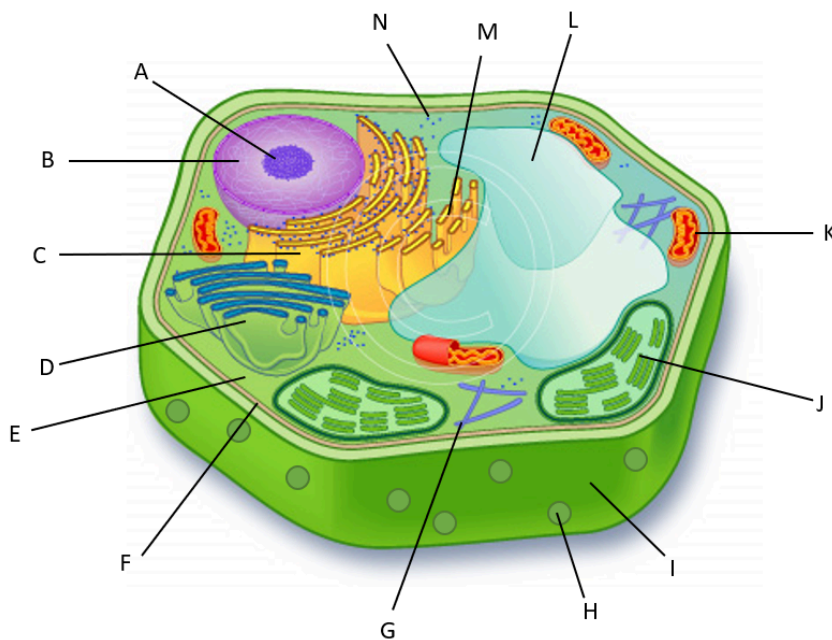
1. What is the function and location (plant, animal, bacteria) of all of the organelles listed below:
  - a. Nucleus
  - b. Chloroplasts
  - c. Mitochondria
  - d. Rough & Smooth ER
  - e. Cytoplasm
  - f. Ribosomes
  - g. Cell Membrane
  - h. Golgi Apparatus
  - i. Lysosomes
  - j. Vacuoles
  - k. Cytoskeleton (including microtubules and microfilaments)
  - l. Centrioles
  - m. Nucleolus
  - n. Nuclear Membrane
  - o. Nuclear Pores
  - p. Cell Wall
  - q. Plasmodesmata
  - r. Gap Junctions
  - s. Microfilaments
  - t. Microtubules
  - u. Tight junctions
  - v. Nucleoid region
2. Who are the 5 cell scientists, and what did they do?
3. What are the 3 parts of the Cell Theory?
4. What are the differences between plant and animal cells? Draw a Venn diagram.
5. What are the differences between prokaryotes and eukaryotes? Give examples of each type of cell.
6. What organelles are involved with protein synthesis?
7. What is the endomembrane system, and what organelles are involved?
8. Compare prokaryotes and viruses; what is the difference? Why are viruses not considered living?

9. Label the bacteria, plant, and animal cell!



### Word Bank

Nucleoid Region  
Cell Wall  
Capsule  
Cell Membrane  
Cytoplasm  
Ribosomes  
Plasmid  
Flagella  
Cilia

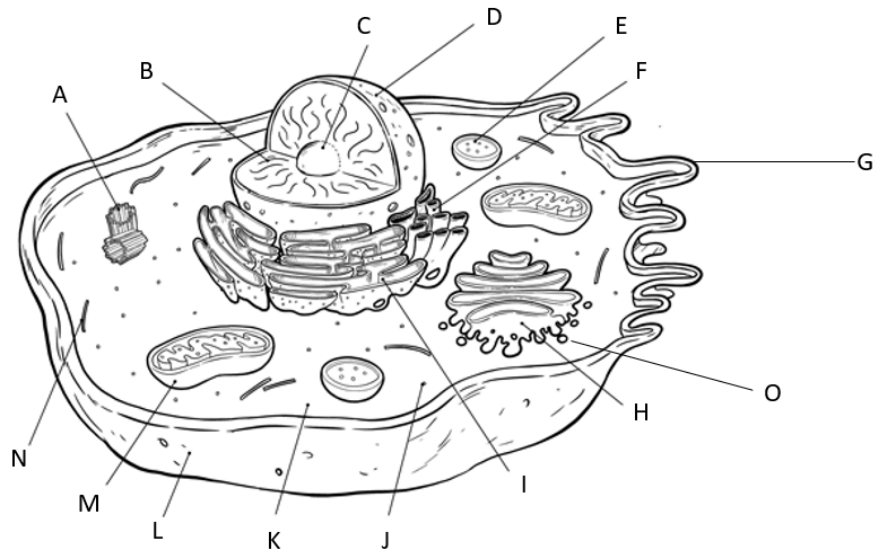


### Word Bank

Central vacuole  
Cell Wall  
Cell Membrane  
Cytoplasm  
Plasmodesmata  
Mitochondria  
Chloroplast  
Rough ER  
Smooth ER  
Ribosomes  
Cytoskeleton  
Golgi body  
Nucleus  
Nucleolus

### Word Bank

Cell Membrane  
Cytoplasm  
Mitochondria  
Rough ER  
Smooth ER  
Ribosomes  
Cytoskeleton  
Golgi body  
Nucleus  
Nucleolus  
Nuclear envelope  
Centrioles  
Cilia  
Vesicles  
Lysosomes



| Organelles/structure          | Function | Collaboration | Cease to collaborate |
|-------------------------------|----------|---------------|----------------------|
| Mitochondria and chloroplast  |          |               |                      |
| Cell wall and central vacuole |          |               |                      |
| Ribosome and RER              |          |               |                      |
| RER and Golgi                 |          |               |                      |
| Golgi vesicles and vacuoles   |          |               |                      |